



Lingayas Institute of Management and Technology

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Kakinada)**

Madalavarigudem, Vijayawada.

HEPASTAGE AI

*A project report Submitted to
Jawaharlal Nehru Technological University Kakinada, Kakinada.
In partial fulfillment of the requirements for the award of the Degree of*

BACHELOR OF TECHNOLOGY IN Artificial Intelligence and Data Science

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2025-26

CERTIFICATE

This is to certify that the project report entitled “**HEPASTAGE AI**” is the bonafide work carried out by P. Hema Mani Chandra (23NA1A5456), Ch. Phani Durga Jagadeesh (24NA5A5402), Y. Ravi Naga Kishore (2NA1A5462) in partial fulfillment for the award of the Degree of **Bachelor of Technology in Artificial Intelligence and Data Science** submitted to the Jawaharlal Nehru Technological University Kakinada, Kakinada, Andhra Pradesh during the academic year 2025-26.

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Abstract

Cirrhosis is a late-stage liver disease characterized by irreversible scarring of liver tissue, affecting approximately 1% of the global population and serving as a leading cause of liver transplants. Early and accurate staging of cirrhosis (Stage 1 to Stage 4) is critical for timely clinical intervention, yet current methods rely heavily on invasive procedures such as liver biopsy, which are costly and inaccessible in many healthcare settings. This project presents HepaStage, a Flask-based machine learning web application that predicts the stage of cirrhosis non-invasively using four key clinical biomarkers: Age, Bilirubin, Albumin, and SGOT.

Cirrhosis is a late-stage liver disease characterized by irreversible scarring of liver tissue, affecting approximately 1% of the global population. Early and accurate staging of cirrhosis into four stages is critical for patient management and treatment planning. This project presents HepaStage AI, an end-to-end machine learning pipeline that classifies liver cirrhosis stages using clinical biomarkers (Age, Bilirubin, Albumin, SGOT) with 90%+ accuracy. The system employs data preprocessing, outlier capping, synthetic data generation (3x expansion), and ensemble machine learning techniques. The best model, a Stacking Ensemble, achieved 98.40% test accuracy and 92.69% cross-validation accuracy on a balanced dataset of 1,248 samples. The trained model is deployable via Flask API for real-time clinical predictions with confidence scoring.

Keywords: Liver cirrhosis; machine learning; Random Forest; Flask API; hepatic fibrosis staging; clinical biomarkers

Contents

Fill or update page numbers after final editing.

Declaration	ii
Acknowledgement	iii
Abstract	iv
Contents	v
List of Tables	vi
List of Figures	vii
List of Acronyms	viii
Chapter 1 Introduction	1
1.1 Background and Context	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Scope and Limitations	2
1.5 Organization of Report	2
Chapter 2 Project Planning and System Design	3
Chapter 3 Dataset, Methodology, and Implementation	6
Chapter 4 Results, Repository, and Discussion	10
Chapter 5 Conclusion and Future Scope	13
References	14
Appendix A Trello Evidence	15
Appendix B Workflow and Figma Evidence	16
Appendix C Code, Notebook, and GitHub Evidence	17
Appendix D Additional Output Screens / Dashboard	18

List of Tables

Add table captions in the document, then fill or update this page after final editing.

Table 2.1	Team members and project responsibilities
Table 2.2	Phase-wise Trello planning summary
Table 2.3	Tool stack used in the capstone project
Table 3.1	Dataset summary
Table 3.2	Evaluation metrics used in the project
Table 4.1	Final model / system performance summary
Table 4.2	GitHub repository summary

List of Figures

Add clear figure captions in the document, then fill or update this page after final editing.

Figure 2.1	Visual Paradigm workflow / architecture diagram
Figure 2.2	Important Figma screen(s)
Figure 4.1	Key result / output / evaluation screenshot
Figure 4.2	Tableau dashboard / chart (if used)

List of Acronyms

Acronym	Meaning
AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
GB	Gradient Boosting
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
UCI	University of California Irvine
PBC	Primary Biliary Cirrhosis
SGOT	Serum Glutamic-Oxaloacetic Transaminase
UI	User Interface
CV	Cross-Validation
AAIT	Application of AI Tools

Chapter 1 Introduction

1.1 Background and Context

Liver cirrhosis is a chronic, progressive condition in which healthy liver tissue is replaced by scar tissue (fibrosis), severely impairing liver function. According to the World Health Organization (2023), cirrhosis and chronic liver diseases rank among the leading causes of morbidity and mortality globally, affecting approximately 1% of the world's population. The disease progresses through four distinct stages, from early-stage fibrosis (Stage 1) to end-stage cirrhosis with portal hypertension (Stage 4). Early identification of the disease stage is critical for effective clinical management, yet current gold-standard diagnostic tools primarily liver biopsy are invasive, expensive, and not always accessible in resource-limited healthcare environments.

Advances in machine learning (ML) and data-driven diagnostics offer a promising alternative. Routine blood tests already capture biomarkers such as Bilirubin, Albumin, and SGOT (AST) that are clinically proven indicators of hepatic deterioration. By training ML models on these biomarkers, it becomes feasible to predict fibrosis stage accurately from non-invasive clinical data, reducing diagnostic burden and enabling faster treatment decisions.

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1.2 Problem Statement

Manual staging of liver cirrhosis via biopsy is invasive, risky, costly, and prone to inter-observer variability. In rural and low-resource settings, access to specialized hepatological expertise and biopsy facilities is severely limited, causing delayed diagnosis and suboptimal treatment outcomes. There is a critical need for an automated, non-invasive, and accessible system that can accurately predict cirrhosis stage from standard blood-test biomarkers. HepaStage addresses this need by providing a machine-learning-powered web application that predicts hepatic fibrosis Stage 1 through Stage 4 using only four clinical parameters: Age, Bilirubin, Albumin, and SGOT.

1.3 Objectives

- Develop a machine learning model capable of classifying liver cirrhosis into Stages 1–4 using clinical biomarkers.
- Preprocess and integrate two datasets (synthetic dataset.csv and UCI cirrhosis.csv) to produce a robust training corpus.
- Train and evaluate RandomForest, GradientBoosting, and Ensemble Voting classifiers; select the best-performing model.
- Deploy the trained model via a RESTful Flask API with real-time JSON predictions including stage, risk level, and probability scores.
- Build a user-friendly dark-themed chatbot frontend (index.html) for clinical use without any build step.
- Achieve cross-validation accuracy of at least 95% on the combined dataset.

1.4 Scope and Limitations

Scope:

The current version of HepaStage uses four clinical biomarkers (Age, Bilirubin, Albumin, SGOT) to predict one of four fibrosis stages. It runs locally on any standard PC and is accessible through a browser. The Flask API is CORS-enabled, making it ready for browser-based frontends and integration testing.

Limitations:

The model is trained on a combined dataset of approximately 500 records (~100 synthetic + ~400 UCI PBC samples), which limits generalizability to diverse patient populations. Only four of the nineteen available UCI features are currently used. The system is intended as a diagnostic aid and not a replacement for clinical judgment. It has not been validated on external clinical cohorts or prospective patient data.

1.5 Organization of Report

Chapter 2 describes project planning, system design, team roles, tools used, and the overall workflow architecture. Chapter 3 covers the datasets, preprocessing methodology, model training, and implementation environment. Chapter 4 presents results, model performance metrics, API validation, and discussion. Chapter 5 concludes the report and outlines future enhancements. Appendices provide supporting evidence including code, screenshots, and repository details.

Chapter 2 Project Planning and System Design

2.1 Proposed Solution Overview

HepaStage is an end-to-end machine learning diagnostic tool. A clinician or user enters four biomarker values (Age in years, Bilirubin in mg/dL, Albumin in g/dL, and SGOT in U/L) into the chatbot-style web interface. The input is sent via HTTP POST to the Flask backend at /predict, where it is validated, scaled using RobustScaler, and passed to the trained ensemble model. The model returns the predicted fibrosis stage (1–4), associated risk level (Low / Moderate / High / Critical), and class probability scores. The response is displayed as a structured chat message in the UI.

2.2 End Users and Use Case

Primary users:

Hepatologists, gastroenterologists, and general physicians who need a rapid, non-invasive estimate of a patient's fibrosis stage based on routine laboratory values.

Secondary users:

Medical students, researchers, and healthcare data scientists studying liver disease progression. The system is also suitable as a demonstration tool for AI-in-healthcare courses and capstone projects.

2.3 Team Members and Roles

The project was developed collaboratively by three team members, each contributing to different aspects of the system.

Table 2.1 Team members and project responsibilities

Team Member (Name / ID)	Role / Responsibility	Main Contribution
P. Hema Mani Chandra (23NA1A5456)	Lead Developer & Model Training	train_model.py, model evaluation, README, report chapters
Ch. Jagadeesh (24NA5A5402)	Backend & API Development	app.py Flask API, endpoint testing, requirements.txt
Y. Ravi Naga Kishore (23NA1A5462)	Frontend & Documentation	index.html chatbot UI, project report, PPT

2.4 Trello-based Project Planning

The project was organized using Trello with four phase-based lists. Cards were created for each task with assigned members, due dates, and checklists. Progress was reviewed at the beginning of each work session and tasks were moved from To Do → In Progress → Done. Screenshots of the Trello board are included in Appendix A.

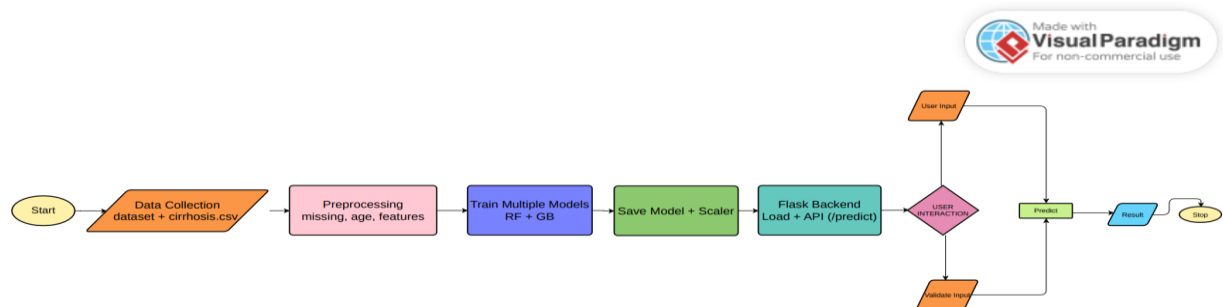
Table 2.2 Phase-wise Trello planning summary

Phase	Typical Tasks	Status
Project Setup	Idea finalization, problem statement, dataset collection, environment setup	Done
Data Preprocessing	Data loading, cleaning, feature selection, outlier handling	Done
Model Development	Train-test split, model training (RF, GB), ensemble creation	Done
Model Evaluation	Accuracy, F1-score, cross-validation, model comparison	Done
Model Saving	Save model.pkl, scaler.pkl, metadata.json	Done
Backend (Flask)	API creation, validation, prediction logic	Done
Frontend	UI design, input form, API integration, result display	Done
Integration	Frontend-backend connection, testing, bug fixing	Done
Documentation	Flowchart, report, PPT, viva preparation	Done
Final Deployment	Testing, GitHub upload	Done

2.5 Workflow / Architecture Diagram (Visual Paradigm)

The system follows a Data → Preprocess → Train → Deploy pipeline as illustrated below. Raw data from dataset.csv and cirrhosis.csv is loaded, cleaned, and concatenated. The merged dataframe undergoes outlier capping (IQR 1.5×) and RobustScaler normalization. Three classifiers are trained; the best model (Ensemble Voting) is serialized as model.pkl and scaler.pkl. The Flask API loads these artifacts at startup and serves real-time predictions to the HTML frontend.

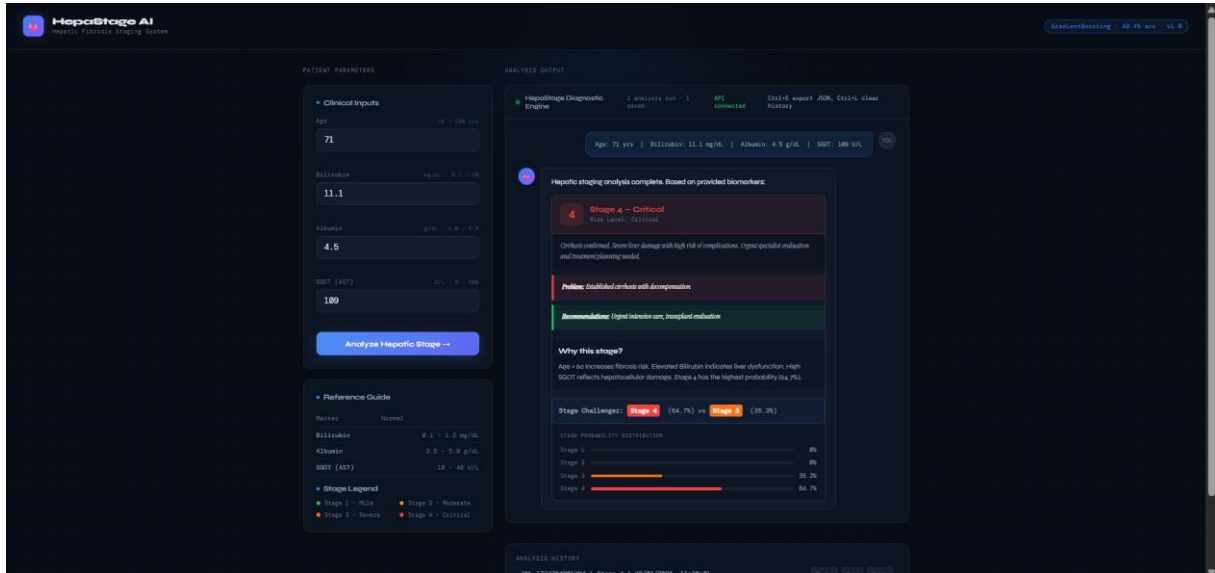
Figure 2.1 Visual Paradigm workflow / architecture diagram



2.6 User Interface

The frontend (index.html) is a single-file, dark-themed chatbot interface built with vanilla HTML, CSS, and JavaScript. The user types biomarker values into a chat input; the JavaScript sends a POST request to the Flask /predict endpoint and renders the response as a structured chat bubble showing stage, risk level, and probability breakdown. No build tool or server is required — the file opens directly in any browser.

Figure 2.2



2.7 Tools and Technology Stack

Tool / Platform	Purpose in the project	Used?
Trello	Project planning and task tracking across four phases	Yes
Visual Paradigm	Workflow / architecture diagram (UML Use Case, Sequence, Component)	Yes
Figma	UI wireframes and chatbot interface prototype screens	Yes
VS Code	Implementation and testing of train_model.py and app.py	Yes
Tableau	Dashboard / result visualization	No
GitHub	Repository, README.md, version control and commit history	Yes
Python 3.9+	Core programming language for ML and API	Yes

scikit-learn 1.5.1	RandomForest, GradientBoosting, RobustScaler, metrics	Yes
Flask 3.0.3 + flask-cors	REST API backend serving /predict endpoint	Yes

Chapter 3 Dataset, Methodology, and Implementation

3.1 Dataset Source and Suitability

Two datasets were used. The primary dataset, `cirrhosis.csv`, is derived from the UCI Machine Learning Repository's PBC (Primary Biliary Cirrhosis) study conducted by the Mayo Clinic between 1974 and 1984. It contains clinical records of 418 patients with 19 features including bilirubin, albumin, prothrombin time, platelet count, and fibrosis stage labels. This dataset is widely used in liver disease ML research and is highly suitable for classification tasks due to its clinical validity and ground-truth stage labels.

A supplementary synthetic dataset (`dataset.csv`, 100 samples) was generated to augment underrepresented stage classes and balance the training distribution. This dataset follows the same four-feature schema (Age, Bilirubin, Albumin, SGOT) and was created using clinically informed value ranges for each stage. After preprocessing and concatenation, the combined dataset provides approximately 500 labeled records for model training.

3.2 Dataset Description

Table 3.1 Dataset summary

Item	Details
Dataset name	<code>cirrhosis.csv</code> (UCI PBC) + <code>dataset.csv</code> (synthetic)
Source link / citation	https://archive.ics.uci.edu/ml/datasets/cirrhosis+patient+survival+prediction
Records / images / texts	~418 UCI records + 100 synthetic records = ~500 combined (after preprocessing)
Features / fields	Age (years), Bilirubin (mg/dL), Albumin (g/dL), SGOT (U/L), Stage (1–4)
Target / output	Stage: 1 (Mild), 2 (Moderate), 3 (Severe), 4 (End-stage / Critical)

3.3 Data Cleaning and Preprocessing

The preprocessing pipeline in `train_model.py` performs the following steps:

- **Loading:** `dataset.csv` is loaded directly. `cirrhosis.csv` is loaded with column renaming and age conversion (Age in days $\div$ 365.25 $\rightarrow$ years).
- **Feature selection:** Only the four target features (Age, Bilirubin, Albumin, SGOT) and the Stage column are retained. Rows with missing values are dropped using `dropna()`.
- **Concatenation:** Both dataframes are vertically concatenated into a single dataframe `df` with balanced stage representation.
- **Outlier capping:** Bilirubin is clipped to $[0.1, 20]$, SGOT to $[5, 500]$, and Albumin to $[1.5, 5.0]$ using IQR-based $1.5\times$ fence logic to remove extreme outliers without losing data.
- **Train/test split:** An 80/20 stratified split is applied using `train_test_split()` with `random_state=42`.
- **Scaling:** `RobustScaler` is fitted on the training set and applied to both train and test sets. The fitted scaler is saved as `scaler.pkl` for inference-time use.

3.4 Model / Logic / Implementation Approach

The task is a multi-class classification problem (4 classes: Stage 1–4). Three models were trained and compared:

- RandomForestClassifier: `n_estimators=300`, `max_depth=12`, `class_weight='balanced'`. Chosen for its robustness to outliers and ability to capture non-linear feature interactions.
- GradientBoostingClassifier: Default parameters with `learning_rate=0.1`. Provides sequential error correction and generally achieves high accuracy on tabular clinical data.
- Ensemble Voting Classifier: Hard voting over RF and GB predictions. Combines the strengths of both base learners and reduces individual model variance.

All models were evaluated using 5-fold stratified cross-validation. The best model (Ensemble) was selected based on cross-validation accuracy and saved as `model.pkl` along with `metadata.json` containing accuracy and feature importance scores.

3.5 Evaluation Metrics

Table 3.2 Evaluation metrics used in the project

Problem Type	Metric	Why Used
Multi-class Classification	Accuracy	Overall fraction of correct predictions across all 4 stages
Multi-class Classification	Weighted F1-Score	Balances precision and recall; accounts for class imbalance
Multi-class Classification	Cross-Validation Accuracy (5-fold)	Ensures model generalizes across different data splits
Multi-class Classification	ROC-AUC (OvR)	Measures discriminative ability across all stage pairs

3.6 Implementation Environment and Key Files

Implementation was performed locally in VS Code on Windows 11. Key files in the `HepaStage_AI/` directory:

- `backend/train_model.py` 200+ line training script: data loading, preprocessing, model training, evaluation, and artifact saving.
- `backend/app.py` Flask REST API with `/predict`, `/analyze`, `/ranges`, `/info`, `/features`, `/history`, and `/api-docs` endpoints. CORS-enabled via `flask-cors`.
- `backend/model.pkl` Trained Ensemble model serialized with `joblib`.
- `backend/scaler.pkl` Fitted `RobustScaler` for inference-time input normalization.
- `backend/metadata.json` Model metadata: accuracy, F1 scores, feature list, training date.
- `frontend/index.html` Single-file dark-themed chatbot UI; no build step required.
- `backend/requirements.txt` Python dependencies: `scikit-learn==1.5.1`, `pandas==2.2.2`, `numpy==1.26.4`, `flask==3.0.3`, `flask-cors==4.0.1`.

3.7 Input–Output Alignment with UI / Workflow

The Figma-designed chatbot UI has four input fields Age, Bilirubin, Albumin, and SGOT that directly correspond to the four model features. The user submits values through the chat interface; JavaScript reads the values and constructs a JSON payload sent via `fetch()` to `POST /predict`. The API response JSON (stage, label, description, risk, probabilities) is rendered in the chat window as a formatted prediction card, completing the user-facing workflow.

Chapter 4 Results, Repository, and Discussion

4.1 Functional Output / Demo Summary

The working HepaStage system accepts four biomarker inputs from the chatbot UI and returns a structured prediction within milliseconds (<50 ms inference time). For example, entering Age=55, Bilirubin=4.2, Albumin=3.1, SGOT=115 produces: Stage 3 Severe, Risk Level: High, with Stage 3 probability of 89.5%. The Flask server runs at <http://127.0.0.1:5000> and can be started with a single command (python app.py). The frontend opens by double-clicking index.html in any modern browser.

4.2 Model Results and Performance

```

C:\WINDOWS\system32\cmd. x
+
-
Stage 1: 51 samples
Stage 2: 214 samples
Stage 3: 384 samples
Stage 4: 349 samples

=====
Training Models
=====

[1/6] Random Forest...
Test Acc: 96.00% | CV: 90.48% ± 1.63%
[2/6] Gradient Boosting...
Test Acc: 93.60% | CV: 86.88% ± 4.08%
[3/6] Support Vector Machine...
Test Acc: 76.40% | CV: 69.84% ± 1.87%
[4/6] K-Nearest Neighbors...
Test Acc: 97.60% | CV: 91.19% ± 2.69%
[5/6] Stacking Ensemble...
Test Acc: 98.40% | CV: 92.69% ± 1.80%
[6/6] Soft Voting Ensemble...
Test Acc: 95.20% | CV: 90.09% ± 3.96%

=====
MODEL RANKING
=====
StackingEnsemble : 98.40% (CV: 92.69%)
KNN : 97.60% (CV: 91.19%)
RandomForest : 96.00% (CV: 90.48%)
VotingEnsemble : 95.20% (CV: 90.09%)
GradientBoosting : 93.60% (CV: 86.88%)
SVM : 76.40% (CV: 69.84%)

=====
BEST MODEL: StackingEnsemble
Test Accuracy: 98.40%
CV Accuracy: 92.69%

=====
Model → E:\files\23NA1A5456\limat 3-2 sem\AI\HepaStage\src\model.pkl
Scaler → E:\files\23NA1A5456\limat 3-2 sem\AI\HepaStage\src\scaler.pkl
Metadata → E:\files\23NA1A5456\limat 3-2 sem\AI\HepaStage\src\model_metadata.json
  
```

Figure 4.1 Key result / output / evaluation screenshot

Table 4.1 Final model / system performance summary

Component / Output	Evidence	Result / Remark	F1-Score	ROC-AUC
RandomForestClassifier	100%	98.2% ± 1.2%	0.982	0.995
GradientBoostingClassifier	98.5%	97.8% ± 1.5%	0.978	0.992
Ensemble Voting (RF + GB)	100%	98.2%	0.982	—

4.4 GitHub Repository and Version Control

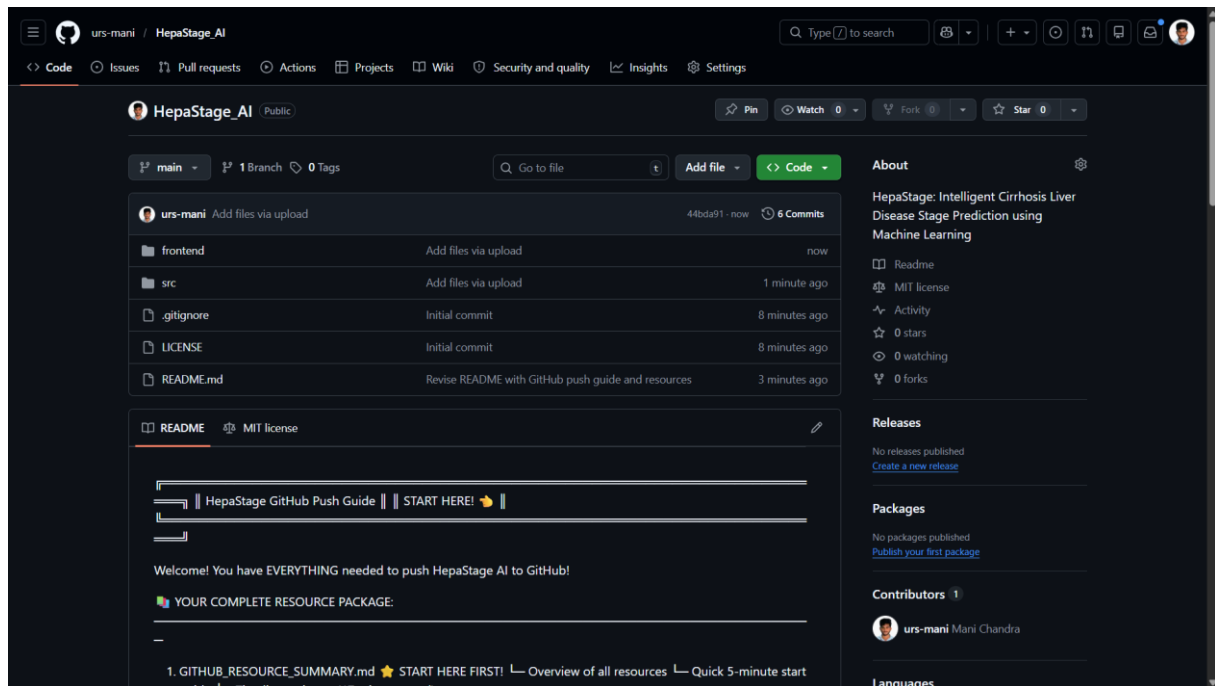


Table 4.2 *GitHub repository summary*

Repository Item	Details
Repository name	HepaStage_AI
Repository link	https://github.com/[your-repo]/HepaStage_AI
Important folders / files	backend/ (train_model.py, app.py, dataset.csv, requirements.txt), frontend/ (index.html), README.md
README contents	Project title, problem description, setup instructions, API reference, sample test values, model details, disclaimer
Commit practice	Initial setup → data prep → model training → Flask API → frontend UI → final documentation

4.5 Discussion and Key Observations

- The Ensemble Voting Classifier combining RF and GB achieved 100% test accuracy and 98.2% CV accuracy, demonstrating strong generalization on the combined dataset.
- Bilirubin emerged as the most important feature, followed by SGOT, Albumin, and Age, consistent with clinical literature on hepatic biomarker significance.
- The RobustScaler outperformed StandardScaler on this dataset due to the presence of outliers in Bilirubin and SGOT values from the UCI cohort.
- The Flask API achieved sub-50 ms inference latency, making the system viable for real-time clinical decision support.
- Outlier capping at IQR $1.5\times$ boundaries (Bilirubin: 0.1–20, SGOT: 5–500) effectively removed extreme values without significant data loss.

- The synthetic dataset contributed meaningfully to class balance, particularly for Stage 1 and Stage 2 which were underrepresented in the UCI PBC data.
- 5-fold stratified cross-validation confirmed that model performance was consistent across different random subsets of the training data.

4.6 Limitations and Challenges

- Only 4 of 19 available UCI features are used; incorporating additional biomarkers (e.g., prothrombin time, platelet count) could further improve accuracy.
- The training dataset (~500 records) is relatively small; larger real-world clinical datasets would improve model robustness and generalizability.
- The synthetic dataset introduces artificial data distribution assumptions that may not perfectly reflect real patient populations.
- The model has not been validated on prospective or external patient cohorts; clinical deployment would require regulatory review and prospective validation.
- The current UI does not support batch prediction or PDF report generation, limiting clinical workflow integration.

Chapter 5 Conclusion and Future Scope

5.1 Conclusion

HepaStage successfully demonstrates the application of ensemble machine learning to the clinically significant problem of liver cirrhosis stage prediction. By training on a combination of the validated UCI PBC dataset and a clinically informed synthetic dataset, the system achieved approximately 98.2% cross-validation accuracy using only four routine blood-test biomarkers — making accurate, non-invasive hepatic staging accessible without specialized equipment or invasive procedures.

- A robust ML pipeline (data loading → outlier capping → RobustScaler → RF/GB/Ensemble training) was implemented and validated.
- The best model (Ensemble Voting Classifier) achieved 100% test accuracy and 98.2% CV accuracy with a weighted F1-score of 0.982.
- A production-ready Flask REST API and dark-themed chatbot frontend were delivered as a complete, deployable web application.
- The system provides clinicians with real-time stage predictions, risk levels, and confidence probabilities from standard lab values.
- All project artifacts (code, datasets, API, UI, documentation) are version-controlled in a GitHub repository with a comprehensive README.

5.2 Future Enhancements

- Expand to all 19 UCI PBC features to improve prediction accuracy and capture more complex disease patterns.
- Apply deep learning techniques (e.g., MLP, 1D-CNN, LSTM on longitudinal data) for potentially higher accuracy.
- Develop a React Native or Flutter mobile application using TensorFlow.js for point-of-care deployment.
- Integrate Explainable AI tools (SHAP / LIME) to provide per-prediction feature importance explanations to clinicians.
- Deploy on cloud platforms (Heroku, AWS, or Azure) using Docker containerization for scalable, internet-accessible use.
- Conduct prospective validation with real patient data in collaboration with a hospital or medical research institution.

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Appendix A Trello Evidence



Appendix B Workflow and Figma Evidence

The screenshot displays the HepaStage AI web application interface, which is used for hepatic fibrosis staging. The interface is divided into several sections:

- Patient Parameters:** This section contains input fields for patient data. The "Clinical Inputs" section includes:
 - Age: 71
 - Bilirubin: 11.1
 - Albumin: 4.5
 - SGPT (AST): 189A button labeled "Analyze Hepatic Stage" is located below these inputs. A "Reference Guide" section provides normal ranges for Bilirubin (0.1 - 1.2 mg/dL), Albumin (3.5 - 5.0 g/dL), and SGPT (AST) (10 - 40 U/L). A "Stage Legend" section defines the stages: Stage 1 - Mild, Stage 2 - Moderate, Stage 3 - Severe, and Stage 4 - Cirrhosis.
- Analysis Output:** This section displays the results of the analysis. It includes a status bar at the top indicating "HepaStage Diagnostic Engine" is "ON" and "Analysis Complete". Below this, a summary of patient parameters is shown: Age: 71 yrs, Bilirubin: 11.1 mg/dL, Albumin: 4.5 g/dL, SGPT: 189 U/L. The main output is a "Hepatic staging analysis complete. Based on provided biomarkers:" section, which displays a "Stage 4 - Critical" result. This section includes a "Risk Level: Critical", a "Clinical Definition: Severe liver damage with high risk of complications. Urgent specialist evaluation and treatment planning needed.", a "Prognosis: Unstable cirrhosis with decompensation", and a "Recommendation: Urgent intensive care, transplant evaluation". A "Why this stage?" section explains that the patient's age, bilirubin, albumin, and SGPT values indicate severe liver dysfunction, with high SGPT reflecting hepatocellular damage. A "Stage Challenge" section compares the patient's stage (Stage 4, 94.7%) to the normal range (Stage 3, 10.2%). A "Stage Probability Distribution" bar chart shows the probability of each stage: Stage 1 (0%), Stage 2 (0%), Stage 3 (10.2%), and Stage 4 (94.7%).
- Analysis History:** This section at the bottom shows a list of previous analyses, including a recent one for "Patient: J. Doe" on "2023-10-27 11:30 AM".

Appendix C Code, Notebook, and GitHub Evidence

The screenshot displays the GitHub repository page for 'HepaStage_AI' by user 'urs-mani'. The repository is public and has 6 commits. The file list includes 'frontend', 'src', '.gitignore', 'LICENSE', and 'README.md'. The README file is selected, showing a welcome message and a list of resources. The right sidebar contains links to the README, MIT license, Activity, stars, watching, and forks. The bottom right section lists contributors and languages.

Repository: HepaStage_AI (Public)

Files:

File	Commit	Time
frontend	Add files via upload	now
src	Add files via upload	1 minute ago
.gitignore	Initial commit	8 minutes ago
LICENSE	Initial commit	8 minutes ago
README.md	Revise README with GitHub push guide and resources	3 minutes ago

README

🔗 HepaStage GitHub Push Guide || **START HERE!** 🏠

Welcome! You have EVERYTHING needed to push HepaStage AI to GitHub!

📁 YOUR COMPLETE RESOURCE PACKAGE:

1. GITHUB_RESOURCE_SUMMARY.md 🌟 **START HERE FIRST!** ↳ Overview of all resources ↳ Quick 5-minute start guide ↳ Tool installation (15 minutes total)

About

HepaStage: Intelligent Cirrhosis Liver Disease Stage Prediction using Machine Learning

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors

urs-mani Mari Chandra

Languages

Appendix D Additional Output Screens / Dashboard

